

2026Fall Quiz

for (MATH230) Probability & Statistics

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Problem 1.1. Four married couples have bought 8 seats in the same row for a concert. In how many different ways can they be seated

- (a) with no restrictions?
- (b) if each couple is to sit together?
- (c) if at least one couple cannot sit next to each other?
- (d) if all the men sit together to the right of all the women?

Note. You can write your answer in calculator-ready form. You don't have to calculate the exact value of fractions or combinations. For example, $8!$, $(4!)^4$, and $\binom{8}{3}/\binom{10}{3}$ are acceptable forms.

Solution.

- (a) With no restrictions, the 8 people can be arranged in

$$\boxed{8!}$$

ways.

- (b) Treat each couple as a single block. The four couple-blocks can be arranged in $4!$ ways, and the two people within each couple can be arranged in $2!$ ways. Hence the number of arrangements is

$$\boxed{4!(2!)^4}.$$

- (c) The complement consists of the arrangements in which all four couples sit together. By parts (a) and (b), the required number is

$$\boxed{8! - 4!(2!)^4}.$$

- (d) The four women must occupy the four leftmost seats and the four men the four rightmost seats. Each group can be arranged independently in $4!$ ways, giving

$$\boxed{4! \cdot 4!}.$$

□

Problem 1.2. A paint-store chain produces and sells latex and semigloss paint. Based on long-range sales, the probability that a customer will purchase latex paint is 0.25. Of those that purchase latex paint, 75% also purchase rollers. But only 25% of semigloss paint buyers purchase rollers. A randomly selected buyer purchases a roller and a can of paint. What is the probability that the paint is latex?

Note. If you introduce symbols (e.g., A , B) in your solution, you must clearly define what each symbol represents.

Solution. Let

$$L = \{\text{the buyer purchases latex paint}\}, \quad S = \{\text{the buyer purchases semigloss paint}\},$$

and

$$R = \{\text{the buyer purchases a roller}\}.$$

Then

$$\mathbb{P}(L) = \frac{1}{4}, \quad \mathbb{P}(R | L) = \frac{3}{4}, \quad \mathbb{P}(R | S) = \frac{1}{4},$$

and, since the paint is either latex or semigloss,

$$\mathbb{P}(S) = 1 - \mathbb{P}(L) = \frac{3}{4}.$$

By the law of total probability,

$$\begin{aligned} \mathbb{P}(R) &= \mathbb{P}(R | L)\mathbb{P}(L) + \mathbb{P}(R | S)\mathbb{P}(S) \\ &= \frac{3}{4} \cdot \frac{1}{4} + \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{8}. \end{aligned}$$

Therefore, by Bayes' rule,

$$\mathbb{P}(L | R) = \frac{\mathbb{P}(R | L)\mathbb{P}(L)}{\mathbb{P}(R)} = \frac{(3/4)(1/4)}{3/8} = \frac{1}{2}.$$

Thus the probability that the paint is latex is 50%.

□

Problem 2.1. For a laboratory assignment, if the equipment is working, the density function of the observed outcome X is

$$f_X(x) = \begin{cases} c(2-x), & 0 < x < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find the value of c so that f_X is a valid density function.
- (b) Find $\mathbb{P}(X < 1)$.
- (c) Given that $X \geq 1$, what is the probability that X will be less than 1.5?

Solution.

- (a) A probability density must integrate to 1. Thus

$$1 = \int_{-\infty}^{\infty} f_X(x) \, dx = \int_0^2 c(2-x) \, dx = 2c,$$

so

$$\boxed{c = \frac{1}{2}}.$$

- (b) Using $c = 1/2$,

$$\boxed{\mathbb{P}(X < 1) = \int_0^1 \frac{1}{2}(2-x) \, dx = \frac{3}{4}}.$$

- (c) By the definition of conditional probability,

$$\mathbb{P}(X < 1.5 \mid X \geq 1) = \frac{\mathbb{P}(1 \leq X < 1.5)}{\mathbb{P}(X \geq 1)}.$$

Here,

$$\mathbb{P}(1 \leq X < 1.5) = \int_1^{3/2} \frac{1}{2}(2-x) \, dx = \frac{3}{16},$$

while

$$\mathbb{P}(X \geq 1) = 1 - \mathbb{P}(X < 1) = \frac{1}{4}.$$

Therefore,

$$\boxed{\mathbb{P}(X < 1.5 \mid X \geq 1) = \frac{3/16}{1/4} = \frac{3}{4}}.$$

□

Problem 2.2. The amount of kerosene, in thousands of liters, in a tank at the beginning of any day is a random amount Y from which a random amount X is sold during that day. Suppose that the tank is not resupplied during the day so that $x \leq y$, and assume that the joint density function of these variables is

$$f_{X,Y}(x, y) = \begin{cases} 2, & 0 < x \leq y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find the marginal density function of X .
- (b) Find the marginal density function of Y .
- (c) Determine if X and Y are independent.
- (d) Find $\mathbb{P}(1/2 < Y < 3/4 \mid X = 1/4)$.

Solution.

- (a) For fixed x with $0 < x < 1$, the support has $x \leq y < 1$. Hence

$$f_X(x) = \int_x^1 f_{X,Y}(x, y) \, dy = \int_x^1 2 \, dy = 2(1 - x).$$

Therefore,

$$f_X(x) = \begin{cases} 2(1 - x), & 0 < x < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (b) For fixed y with $0 < y < 1$, the support has $0 < x \leq y$. Thus

$$f_Y(y) = \int_0^y f_{X,Y}(x, y) \, dx = \int_0^y 2 \, dx = 2y.$$

Therefore,

$$f_Y(y) = \begin{cases} 2y, & 0 < y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (c) If X and Y were independent, then $f_{X,Y}(x, y) = f_X(x)f_Y(y)$ almost everywhere. But for $0 < y < x < 1$,

$$f_{X,Y}(x, y) = 0, \quad f_X(x)f_Y(y) = 4y(1 - x) > 0.$$

Hence

$$X \text{ and } Y \text{ are not independent.}$$

- (d) For $0 < x < 1$, the conditional density of Y given $X = x$ is

$$f_{Y|X}(y \mid x) = \frac{f_{X,Y}(x, y)}{f_X(x)} = \frac{1}{1 - x}, \quad x < y < 1.$$

When $X = 1/4$,

$$f_{Y|X}(y \mid 1/4) = \frac{4}{3}, \quad \frac{1}{4} < y < 1.$$

Therefore,

$$\mathbb{P}\left(\frac{1}{2} < Y < \frac{3}{4} \mid X = \frac{1}{4}\right) = \int_{1/2}^{3/4} \frac{4}{3} \, dy = \frac{1}{3}.$$

□

Problem 3.1. The probability distribution of the discrete random variable X is

$$p_X(x) = \binom{5}{x} \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{5-x}, \quad x = 0, 1, 2, 3, 4, 5.$$

Find the mean of X .

Solution. By the definition of expectation for a discrete random variable,

$$\begin{aligned} \mathbb{E}[X] &= \sum_{x=0}^5 x p_X(x) \\ &= \sum_{x=0}^5 x \binom{5}{x} \left(\frac{1}{5}\right)^x \left(\frac{4}{5}\right)^{5-x} \\ &= \frac{1280 + 1280 + 480 + 80 + 5}{3125} \\ &= \boxed{1}. \end{aligned}$$

□

Problem 3.2. Let X and Y be random variables with joint density function

$$f_{X,Y}(x, y) = \begin{cases} 4xy, & 0 < x, y < 1, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find the expected value of $X + Y$.
- (b) Find the expected value of XY .

Solution.

- (a) By the definition of expectation for a function of jointly continuous random variables,

$$\begin{aligned} \mathbb{E}[X + Y] &= \int_0^1 \int_0^1 (x + y) f_{X,Y}(x, y) \, dy \, dx \\ &= \int_0^1 \int_0^1 (4x^2y + 4xy^2) \, dy \, dx \\ &= \int_0^1 \left(2x^2 + \frac{4}{3}x\right) \, dx \\ &= \boxed{\frac{4}{3}}. \end{aligned}$$

- (b) Again using the definition of expectation,

$$\begin{aligned} \mathbb{E}[XY] &= \int_0^1 \int_0^1 xy f_{X,Y}(x, y) \, dy \, dx \\ &= 4 \int_0^1 \int_0^1 x^2 y^2 \, dy \, dx \\ &= 4 \left(\int_0^1 x^2 \, dx \right) \left(\int_0^1 y^2 \, dy \right) \\ &= \boxed{\frac{4}{9}}. \end{aligned}$$

□

Problem 4.1. From a sack of fruit containing 2 oranges, 1 apple, and 2 bananas, a random sample of 3 pieces of fruit is selected. If X is the number of oranges and Y is the number of apples in the sample, find the covariance of the random variables X and Y .

Solution. There are $\binom{5}{3} = 10$ equally likely samples. To obtain $X = x$ oranges and $Y = y$ apples, the sample must contain $3 - x - y$ bananas. Hence the joint probability distribution is

$$p_{X,Y}(x, y) = \frac{\binom{2}{x} \binom{1}{y} \binom{2}{3-x-y}}{\binom{5}{3}}$$

for the admissible values of x and y , giving

$p_{X,Y}(x,y)$		x		
		0	1	2
y				
0		0	$\frac{2}{10}$	$\frac{2}{10}$
1		$\frac{1}{10}$	$\frac{4}{10}$	$\frac{1}{10}$

The marginal probabilities give

$$\begin{aligned}\mathbb{E}[X] &= (1) \left(\frac{2+4}{10} \right) + (2) \left(\frac{2+1}{10} \right) = \frac{6}{5}, \\ \mathbb{E}[Y] &= (1) \left(\frac{1+4+1}{10} \right) = \frac{3}{5}.\end{aligned}$$

Also, only the pairs $(1, 1)$ and $(2, 1)$ contribute to $\mathbb{E}[XY]$:

$$\begin{aligned}\mathbb{E}[XY] &= \sum_x \sum_y xy p_{X,Y}(x, y) \\ &= (1)(1) \frac{4}{10} + (2)(1) \frac{1}{10} = \frac{3}{5}.\end{aligned}$$

Therefore, by the covariance formula,

$$\boxed{\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = \frac{3}{5} - \left(\frac{6}{5}\right)\left(\frac{3}{5}\right) = -\frac{3}{25}}.$$

□

Problem 4.2. Let X and Y be random variables with joint density function

$$f_{X,Y}(x,y) = \begin{cases} \frac{1}{2}, & 0 < x \leq y < 2, \\ 0, & \text{elsewhere.} \end{cases}$$

- (a) Find $\text{Var}(X)$ and $\text{Var}(Y)$.
- (b) Find $\text{Cov}(X, Y)$.
- (c) Find $\rho_{X,Y}$, the correlation coefficient between X and Y .
- (d) Find $\text{Var}(X + Y)$.

Hint: You may use $\mathbb{E}[X] = \frac{2}{3}$, $\mathbb{E}[Y] = \frac{4}{3}$, $\mathbb{E}[X^2] = \frac{2}{3}$, and $\mathbb{E}[Y^2] = 2$.

Solution. The support is the triangular region $0 < x \leq y < 2$.

- (a) Using the given moments and the variance formula,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \frac{2}{3} - \left(\frac{2}{3}\right)^2 = \boxed{\frac{2}{9}},$$

$$\text{Var}(Y) = \mathbb{E}[Y^2] - (\mathbb{E}[Y])^2 = 2 - \left(\frac{4}{3}\right)^2 = \boxed{\frac{2}{9}}.$$

- (b) We first calculate the mixed moment using the joint density:

$$\begin{aligned} \mathbb{E}[XY] &= \int_0^2 \int_0^y xy f_{X,Y}(x,y) \, dx \, dy \\ &= \int_0^2 \int_0^y \frac{xy}{2} \, dx \, dy \\ &= \frac{1}{4} \int_0^2 y^3 \, dy = 1. \end{aligned}$$

Therefore,

$$\boxed{\text{Cov}(X, Y) = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] = 1 - \left(\frac{2}{3}\right)\left(\frac{4}{3}\right) = \frac{1}{9}}.$$

- (c) Since both variances are positive, the correlation coefficient is

$$\boxed{\rho_{X,Y} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{1/9}{\sqrt{(2/9)(2/9)}} = \frac{1}{2}}.$$

- (d) Using the variance of a sum,

$$\begin{aligned} \text{Var}(X + Y) &= \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y) \\ &= \frac{2}{9} + \frac{2}{9} + 2\left(\frac{1}{9}\right) = \boxed{\frac{2}{3}}. \end{aligned}$$

□